LessonPlan Career Roadmap

Science - PCM (Physics, Chemistry, Mathematics): Class 6 to Career

STATUS: DRAFT FOR HUMAN REVIEW - NOT FOR PUBLIC USE

Drafted and web-checked on 17 Oct 2026 by an AI assistant. Sources were mostly coaching-institute pages, scholarship aggregators and news reports; the HBCSE Maths Olympiad brochure was the only official document found. NTA, UPSC, KEA and DST sites could not be opened directly. Items marked **[VERIFY]** are unconfirmed or conflicting and must be checked on the official site before publishing. Unmarked items are supported by at least one secondary source but should still be spot-checked. See section 9 for the verification log.

1. Pathway overview

Stage	When	Key decision	What it unlocks
Foundation	Class 6-8	Start Maths and Science Olympiads; build problem-solving habit	Olympiad ladder, NMMS scholarship (Class 8)
Pre-board	Class 9-10	Take Standard Maths (CBSE) and Science; start IOQM (from Class 8) and Junior Science Olympiad	Keeps PCM open; Basic Maths may block Maths in Class 11 **[VERIFY with school / CBSE]**
Stream choice	After Class 10	Choose PCM (+ optional Computer Science / Biology / Economics) in Class 11-12 (PUC in Karnataka)	Eligibility for JEE, BITSAT, KCET, NDA, B.Arch, research tracks
Preparation	Class 11-12	Boards + JEE Main / Advanced / KCET / BITSAT; Olympiads (NSE, INMO)	IIT, NIT, IIIT, BITS, state and private engineering seats
Degree	After Class 12	Engineering, architecture, pure science, data / computing, defence, aviation	Professional or research career
Postgraduate	3-6 years later	MTech, MS, MSc, MBA, PhD	Specialist and research roles
Career	-	Engineering, technology, research, defence, architecture, finance	See section 7

2. Stage-by-stage nodes

Class 6-8: Foundation

- Decision: join Maths and Science Olympiads for exposure. Not mandatory, but it builds the profile and the habit used later.
- IOQM (Indian Olympiad Qualifier in Mathematics) is open from Class 8 (see section 3).
- Unlocks: NMMS scholarship exam in Class 8; INSPIRE-MANAK idea awards (Class 6-12).

Class 9-10: Pre-board

- Decision: in CBSE, choose Standard Maths for Class 10 if there is any chance of PCM, engineering or architecture. Basic Maths may restrict Maths in Class 11. **[VERIFY current CBSE rule and school policy]**
- Junior Science Olympiad (NSEJS) is for Class 10 and below.
- Prerequisite for PCM: strong Class 10 Maths and Science; algebra, geometry and basic physics concepts.

After Class 10: Stream choice

- Choose PCM. Add one optional subject: Computer Science (software, data), Biology (PCMB, keeps NEET open), Economics, or Physical Education.
- Karnataka note: Class 11-12 is the 2-year Pre-University Course (PUC) after SSLC; CBSE/ICSE schools run Class 11-12 directly.

- Engineering via KCET needs Physics, Chemistry and Maths in Class 12; NDA Air Force and Navy wings also need Physics and Maths.

Class 11-12: Preparation

- Board marks matter: KCET rank uses 50% of PUC marks, and many central institutes ask for a minimum Class 12 percentage or percentile. **[VERIFY JEE-based admission rule for 2026-27]**
- Apply in parallel: JEE Main (two sessions), JEE Advanced (if qualified), KCET, BITSAT, COMEDK, plus NDA and other options if interested.
- Olympiad track for strong students: NSE (Physics / Chemistry / Astronomy) in Class 10-12; IOQM -> RMO -> INMO for Maths.

After Class 12: Degree

- Engineering and technology: BE / BTech through JEE, KCET, BITSAT, COMEDK and private university exams.
- Architecture: B.Arch through NATA or JEE Main Paper 2A.
- Pure science and mathematics: BSc, BS-MS (IISER / NISER / IISc), CMI, ISI (B.Stat / B.Math).
- Defence and aviation: NDA, technical entries, pilot and merchant navy routes.

3. Olympiads and early exams (Class 6-10 and beyond)

Exam	Conducted by	Classes	Levels	Timing (2026-27 cycle)
IOQM (Maths)	HBCSE with MTAI	8-12 (born 1 Aug 2007 to 31 Jul 2014)	IOQM, then RMO, then INMO, then IMO training camp and IMO selection **[VERIFY post-INMO steps]**	IOQM held 6 Sep 2026; RMO likely 15 Nov 2026; INMO likely 17 Jan 2027 **[VERIFY RMO/INMO dates]**
NSE: NSEP, NSEC, NSEA (Physics, Chemistry, Astronomy)	IAPT	10-12 (born 1 Jul 2007 to 30 Jun 2012)	Stage 1 NSE, then HBCSE national olympiads (INPhO, INChO, INAO) **[VERIFY]**	Stage 1: NSEA 21 Nov, NSEP / NSEC 22 Nov 2026; INPhO about Jan-Feb 2027. Registration closed 14 Sep
NSEJS then INJSO (Junior Science)	IAPT then HBCSE	10 and below (age rule applies)	Stage 1 NSEJS, Stage 2 INJSO **[VERIFY]**	Nov, then Jan-Feb **[VERIFY]**
National Science Olympiad (NSO) / IMO	Science Olympiad Foundation (SOF)	1-12	Level 1 (school) then Level 2 **[VERIFY]**	Level 1 Oct-Dec, Level 2 Jan-Mar **[VERIFY]**
INSPIRE-MANAK	DST with NIF	6-12 (Science stream for Class 11-12)	School nominates up to 5 ideas; district, state, national	Last nominations closed 15 Sep 2025; check inspireawards-dst.gov.in
NMMS (scholarship test)	Karnataka DSERT / KSEAB	8 (govt, aided and local body schools)	Single written test	Apply Sep-Oct; exam Jan-Feb (last held 1 Feb 2026)
NTSE	NCERT; Karnataka Stage 1 via DSERT	10	Stage 1 state, Stage 2 national	No sign of discontinuation; 2026-27 notification expected Aug-Oct **[VERIFY]**

Tip for parents: Olympiad performance can lead to direct benefits later. For example, CMI exempts outstanding national olympiad performers from its entrance test. **[VERIFY exact rule]**

4. Entrance exams after Class 10 / 12

JEE Main (Joint Entrance Examination - Main)	
Conducting body	National Testing Agency (NTA); jeemain.nta.nic.in
Used for	NITs, IIITs, other centrally funded technical institutes, many state and private colleges; also the qualifier for JEE Advanced
Eligibility to appear	Class 12 passed in 2024 or 2025, or appearing in 2026. No minimum percentage and no age limit to apply. Admission to NITs / IIITs additionally needs about 75% in Class 12 (65% for SC/ST) or top 20 percentile in the board **[VERIFY 2026-27 rule]**

Attempts	Up to three consecutive years is the usual rule **[VERIFY]**
Frequency	Two sessions a year; best score counts. 2026: Session 1 on 21-30 January, Session 2 on 2-9 April (scheduled dates) **[VERIFY]**
Pattern (Paper 1: BE / BTech)	75 questions, 300 marks, 3 hours, computer-based. 25 questions per subject: 20 MCQs + 5 numerical value questions. Reported as +4 / -1 for MCQs. Whether numerical questions carry negative marking and whether all 5 are compulsory is contested **[VERIFY in 2026 bulletin]**
Paper 2	Paper 2A (B.Arch) and Paper 2B (B.Planning): 400 marks each **[VERIFY]**
Languages	English, Hindi and several regional languages **[VERIFY Kannada]**

JEE Advanced	
Conducting body	Rotates among the seven IITs; IIT Roorkee in 2026 (jeeadv.ac.in)
Used for	IITs (about 23 institutes) and related institutes via JoSAA counselling **[VERIFY]**
Eligibility	Top 2,50,000 qualifiers of JEE Main (BE / BTech); born on or after 1 Oct 2001 (5-year relaxation for SC/ST/PwD); Class 12 first appeared no earlier than the previous year; at most 2 attempts in 2 consecutive years
Frequency / date	Once a year; 2026 exam on Sunday 17 May 2026 (registration 23 Apr - 2 May); results 1 June
Pattern	Computer-based; two compulsory papers of 3 hours each (Paper 1 morning, Paper 2 afternoon); Physics, Chemistry, Mathematics. Marking varies by question type and changes yearly **[VERIFY]**

KCET (Karnataka Common Entrance Test) - Engineering	
Conducting body	Karnataka Examinations Authority (KEA); kea.kar.nic.in and cetonline.karnataka.gov.in
Used for	Engineering, architecture (with NATA), B.Pharm, farm sciences and other Karnataka professional courses
Minimum qualification	2nd PUC / Class 12 with Physics, Chemistry, Maths; at least 45% aggregate in PCM for General, 40% for reserved categories. One source says 50% for General **[VERIFY in KEA brochure]**
Frequency / date	Once a year; 2026 exam 23-24 April (Kannada language test on 22 April for eligible candidates); results 6 June 2026
Pattern	Three papers for a PCM candidate: Physics, Chemistry and Maths, 60 MCQs each (180 in total), 1 mark each, 80 minutes per paper. Offline OMR, English and Kannada.
Marking	No negative marking
Rank basis	50% KCET score + 50% 2nd PUC marks

BITSAT (BITS Pilani)	
Conducting body	BITS Pilani (Pilani, Goa, Hyderabad campuses); admissions.bits-pilani.ac.in
Eligibility	Class 12 with Physics, Chemistry and Maths and English; aggregate and per-subject minimums apply (commonly quoted: 75% aggregate in PCM and 60% in each subject). Sources garbled, check brochure **[VERIFY]**
Frequency / date	Two sessions; 2026: Session 1 on 15-17 April, Session 2 on 24-26 May. Best score counts. Registration opens December **[VERIFY]**
Pattern	Computer-based, 3 hours, 130 MCQs: Physics, Chemistry, English Proficiency, Logical Reasoning and Mathematics (or Biology for B.Pharm)
Marking	+3 correct, -1 wrong

COMEDK UGET

Conducting body	COMEDK (Karnataka private engineering colleges); comedk.org
Eligibility	Second PUC / Class 12 with Physics, Chemistry, Maths and English; at least 45% aggregate for General **[VERIFY reserved-category minimum]**
Frequency / date	Once a year; 2026: 9 May in CBT mode, results 26 May
Pattern	180 marks: Physics 60, Chemistry 60, Maths 60 at 1 mark each (so about 60 questions per subject) **[VERIFY question count]**
Marking	No negative marking

NDA and NA (National Defence Academy / Naval Academy) - UPSC	
Conducting body	Union Public Service Commission (UPSC); upsc.gov.in
Eligibility	Class 12 pass (or appearing). Army wing: any stream. Air Force and Naval wings, and Naval Academy: Physics and Maths required. Unmarried; about 16.5 to 19.5 years of age (birth-date window changes each cycle) **[VERIFY]**
Frequency / date	Twice a year (NDA I and NDA II); NDA II 2026 written exam was on 13 September 2026. Next cycle: NDA I 2027 notification expected around December **[VERIFY]**
Pattern	Written test of 900 marks: Mathematics 300 marks and General Ability Test 600 marks, 2.5 hours each. SSB interview adds 900 marks (total 1800)
Marking	Negative marking of one-third of the question marks for wrong answers **[VERIFY exact values]**

NATA (National Aptitude Test in Architecture)	
Conducting body	Council of Architecture (CoA); nata.in
Used for	B.Arch admissions (JEE Main Paper 2A is the alternative)
Eligibility	Passed or appearing in Class 12 with Physics and Maths. CoA minimum for B.Arch about 45% aggregate; most colleges ask for PCM and a higher percentage **[VERIFY]**
Frequency / dates	Multiple test sessions in one academic year, best score counts. 2026 reported as Phase 1 (Apr-Jun, Fri / Sat) and Phase 2 (7-8 Aug) **[VERIFY on nata.in]**
Pattern	Drawing test (offline) plus an aptitude test (online)

Research-oriented and other routes

Exam / route	For	Key details
IISER Aptitude Test (IAT)	BS-MS (5 yr) at IISERs; BS / BTech at Bhopal and Tirupati	60 MCQs, 15 each (Bio, Chem, Maths, Phy), 3 hr, 240 marks, +4 / -1; need 60% (55% SC/ST/PwD) and at least three of PCMB; 2026 exam on 7 June
NEST	Integrated MSc at NISER / UM-DAE CEBS	Once a year, about June; CBT **[VERIFY details]**
CMI Entrance Exam	BSc (Hons) Maths / CS / Data Science at Chennai Mathematical Institute	2026 test on 2 May; applications closed 4 April; Part A objective (40 marks), Part B descriptive (80 marks); no minimum percentage; next cycle applications about March 2027 **[VERIFY totals]**
ISI Admission Test	B.Stat (Kolkata) and B.Math (Bengaluru) at Indian Statistical Institute	2026 test on 10 May; Maths and English compulsory in Class 12; apply by about March **[VERIFY next cycle]**
CUET-UG	BSc Physics / Maths / Chemistry / CS at central and other universities	50 MCQs per subject, 60 min, +5 / -1, computer-based; month May-June **[VERIFY]**
VITEEE, SRMJEEE and other private university exams	Private engineering universities	Dates and patterns vary by university **[VERIFY]**

Postgraduate and beyond (preview)

Exam	Body	For	Tentative timing
GATE (Engineering / Science)	IIT-organised	MTech, PhD, PSU jobs	Feb **[VERIFY]**
CAT / GMAT and other management tests	IIMs / others	MBA after BTech	CAT: Nov **[VERIFY]**
IIT JAM	IIT-organised	MSc Physics, Chemistry, Maths at IITs	Feb **[VERIFY]**
CSIR-UGC NET	NTA	JRF, lectureship	Twice a year **[VERIFY]**
UPSC CSE / ESE and state PSC exams	UPSC / states	Civil and engineering services	Yearly **[VERIFY]**

5. Scholarships

Scholarship	Provider	Who can apply	Amount	Window
NMMS	Govt. of India via Karnataka DSERT	Class 8 in govt/aided/local body schools; family income up to 3.5 lakh; at least 55% in Class 7 (50% SC/ST) **[VERIFY: single source]**	12,000 per year (1,000 per month), Class 9-12	Apply Sep-Oct; exam Jan-Feb
INSPIRE-MANAK	DST with NIF	Class 6-12, idea nominated by school (max 5 per school)	10,000 one-time	Nominations around Jul-Sep; inspireawards-dst.gov.in
INSPIRE-SHE	DST	Top 1% in Class 12 boards, or top 10,000 in JEE Main / Advanced / NEET, or olympiad scholars, studying BSc / BS / Int. MSc in natural and basic sciences. No income limit.	80,000 per year (60,000 scholarship + 20,000 summer-project mentorship), renewable	Expected deadline 1 Dec 2026, not yet notified; online-inspire.gov.in **[VERIFY]**
CSSS (Central Sector Scheme)	Ministry of Education via National Scholarship Portal	Top 20% of successful candidates in own board and stream; family income up to 4.5 lakh	12,000 per year for UG years 1-3; 20,000 per year for years 4-5 / PG	2026-27 portal opened 1 Jun 2026; one listing shows 31 Oct 2026 deadline **[VERIFY]**
Post-matric scholarships	National Scholarship Portal and Karnataka SSP	SC/ST/OBC/minority categories, income limits	Varies	Typically Jul-Nov **[VERIFY]**
Institute and corporate scholarships	IITs, BITS, private foundations	Merit-cum-means, need based	Varies	Varies; check each institute **[VERIFY]**

Note: KVPY was discontinued and its role absorbed by INSPIRE-SHE; a few listings still mention KVPY scholars. **[VERIFY]**

6. Courses and colleges

Course	Duration	Entry	Note
BE / BTech (Computer Science, AI and Data, Electronics, Mechanical, Civil, Electrical, Chemical, Aerospace and others)	4 yr	JEE Main / Advanced, KCET, COMEDK, BITSAT, private exams	Largest PCM pathway; branch choice depends on rank
B.Arch	5 yr	NATA or JEE Main Paper 2A	Licence from Council of Architecture
BSc Physics / Chemistry / Maths / Computer Science / Statistics / Electronics	3-4 yr	CUET-UG or university	Research and teaching route; MSc and PhD later
BS-MS (IISER / NISER / IISc)	4-5 yr	IAT, NEST, IISc admission	Research focused
BSc (Hons) at CMI; B.Stat / B.Math at ISI	3 yr	CMI entrance; ISI admission test	Maths, statistics, data science; strong research pipeline
BCA / BSc Data Science / Integrated MTech	3-5 yr	CUET, university tests, JEE	Computing and data careers

Course	Duration	Entry	Note
NDA (Army / Navy / Air Force) then BTech / BSc	3 yr at NDA + 1 yr training	NDA exam + SSB	Officer entry, degree funded by services
Pilot training (CPL) / Merchant Navy (BSc Nautical, BTech Marine)	1.5-4 yr	DGCA norms; IMU CET for marine **[VERIFY]**	Physics and Maths needed in Class 12 **[VERIFY]**
B.Des	4 yr	UCEED, NID DAT (any stream)	See section 8

Examples of institutions (illustrative, not a ranking): IITs and IISc, NITK Surathkal, IIIT Bangalore, BITS Pilani, IISERs, ISI Bengaluru, CMI, RV College of Engineering, PES University, BMS College of Engineering, MS Ramaiah. **[VERIFY availability, fees and admission route]**

7. Career outcomes

Path	Job roles	Higher study
Software and computing	Software engineer, data scientist, AI/ML engineer, cybersecurity analyst	MTech / MS, MBA, PhD
Core engineering	Mechanical, civil, electrical, electronics, chemical, aerospace engineer; PSU roles via GATE	MTech, MS, MBA, ESE / UPSC
Architecture and planning	Architect, urban planner, interior and landscape designer	M.Arch, M.Plan
Pure science and research	Physicist, chemist, mathematician, statistician, researcher, professor	MSc, PhD, CSIR / INSPIRE fellowships
Defence and aviation	Army / Navy / Air Force officer, pilot, merchant navy officer	Service training, defence technology courses
Finance and analytics	Quant analyst, actuary, data analyst, economist	MBA, actuarial exams, MSc Statistics **[VERIFY]**

8. Related umbrellas this stream connects to

- Technology / Computer Science: a large share of PCM students; BTech CSE, BCA and data science.
- Design: B.Des through UCEED and NID DAT accept students from any stream; confirm current eligibility. **[VERIFY]**
- Defence: NDA (Army wing any stream; Air Force and Navy need PCM); technical entry schemes for engineering graduates. **[VERIFY]**
- Law: CLAT is open to Science students; patent law and technology law combine both. **[VERIFY]**
- Commerce-adjacent: Economics, Actuarial Science, Management (IPMAT / CAT) welcome strong Maths students. **[VERIFY exam routes]**
- Medicine: PCMB students can keep NEET open; see the Science PCB document.

9. Verification log (17 Oct 2026)

What the web check found. Confidence reflects secondary sources unless noted.

Item	Finding	Confidence
JEE Main pattern	75 Q, 300 marks, 3 hr CBT, 25 per subject (20 MCQ + 5 numerical); +4 / -1 MCQ. Numerical negative marking and compulsory-5 rule conflict across sources	High for structure; Low for numerical rules
JEE Main sessions	Jan 21-30 and Apr 2-9, 2026 (scheduled); best score counts. Not confirmed that both were held as scheduled	Medium
JEE Main eligibility	No minimum % or age limit to appear; admission-stage 75% rule not confirmed for 2026	Medium / Low
JEE Advanced	IIT Roorkee, 17 May 2026, CBT, top 2.5 lakh, two compulsory papers, max 2 attempts, results 1 Jun	High (multiple sources)

Item	Finding	Confidence
KCET engineering	45% PCM (General), 40% reserved; rank 50:50 with PUC; 60 Q per paper; no negative; 23-24 Apr; one source says 50%	Medium-High
BITSAT	130 MCQs, 3 hr, +3 / -1, two sessions (Apr and May 2026); eligibility percentages garbled across sources	High for pattern; Low for eligibility
COMEDK UGET	9 May 2026, CBT, 60 marks each in P / C / M, no negative; 45% General; question count inferred	Medium
NDA	Two papers 300 + 600 = 900, SSB 900, NDA II on 13 Sep 2026, Physics and Maths for Air Force / Navy; negative marking one-third	Medium-High
IOQM / RMO / INMO	From HBCSE brochure (official): Class 8-12, IOQM on 6 Sep 2026; RMO and INMO dates from a secondary source	High for IOQM; Medium for later dates
NSE (IAPT)	NSEA 21 Nov, NSEP / NSEC / NSEB 22 Nov 2026, Class 10-12, offline; later stages expected Jan-Feb 2027	High (IAPT site quoted)
NATA	Dates and number of attempts conflict; CoA minimum quoted as 45%, not 50%	Low
IAT / CMI / ISI	IAT 60 Q, 240 marks, +4 / -1, 7 Jun 2026; CMI test 2 May, apps closed 4 Apr; ISI test 10 May 2026	Medium-High
Scholarships	INSPIRE-SHE 80,000 (60,000 + 20,000), no income cap; CSSS 12,000 / 20,000; NMMS 12,000 per year; MANAK 10,000	Medium

Still to verify on official sites

- JEE Main: 2026 bulletin text on numerical-question marking, attempts, Kannada, and the Class 12 percentage rule for NITs / IITs; JoSAA counselling details.
- BITSAT eligibility percentages; COMEDK question count and reserved-category eligibility; NDA age window and negative-marking values.
- NATA dates and attempts; NEST; CMI total marks; VITEEE and other private exams.
- Post-INMO selection steps, NSEJS and SOF dates, NTSE 2026-27, pilot and merchant navy requirements, CBSE Basic-Maths rule.
- Official sources: jeemain.nta.nic.in, jeeadv.ac.in, josaa.nic.in, kea.kar.nic.in, comedk.org, admissions.bits-pilani.ac.in, upsc.gov.in, nata.in, iiseradmission.in, cmi.ac.in, admission.isical.ac.in, olympiads.hbcse.tifr.res.in, iapt.org.in, online-inspire.gov.in, scholarships.gov.in.
- Dates shift every year. Replace "tentative month" with the official notification date before publishing.

ONE-PAGE GIST: Science - PCM

Draft - verify all **[VERIFY]** items in the detailed version before publishing.

Stage	What to do
Class 6-8	Try Olympiads (IOQM from Class 8, NSO / IMO); Class 8: NMMS scholarship test
Class 9-10	Take Standard Maths; Junior Science Olympiad; build strong Maths and Physics
After Class 10	Choose PCM (+ Computer Science / Biology optional); PUC in Karnataka
Class 11-12	Boards + JEE Main (2 sessions) -> JEE Advanced; also KCET, BITSAT, COMEDK; NSE / INMO for strong students
After Class 12	BE / BTech, B.Arch, BSc / BS-MS (IISER, CMI, ISI), NDA, pilot or merchant navy, BCA / Data Science
Postgraduate	GATE, JAM, CAT, UPSC, CSIR-NET

Exam	When	Pattern	Marking
JEE Main	Jan and Apr sessions	75 Q, 300 marks, 3 hr CBT; 25 per subject	+4 / -1 (numerical: check)
JEE Advanced	May (2026: 17 May)	2 papers, 3 hr each, CBT; top 2.5 lakh of JEE Main	Varies by question
KCET	April (2026: 23-24)	Phy, Chem, Maths: 60 Q each, 80 min each	+1 / 0
BITSAT	Apr and May sessions	130 Q, 3 hr CBT	+3 / -1
COMEDK	May (2026: 9 May)	180 marks: P / C / M 60 each	+1 / 0
NDA	Twice a year	Maths 300 + GAT 600; SSB 900	1/3 negative

Key rules: KCET rank = 50% KCET + 50% PUC marks; KCET needs about 45% PCM (General). JEE Advanced needs JEE Main top 2.5 lakh, max 2 attempts. **[VERIFY]**

Scholarships: NMMS (Class 8, 12,000/yr), INSPIRE-MANAK (Class 6-12, 10,000), INSPIRE-SHE (80,000/yr) and CSSS (12,000/yr) after Class 12, post-matric schemes via NSP / SSP. **[VERIFY income limits and deadlines]**

Careers: engineer (software, core, aerospace), architect, scientist, statistician, defence officer, pilot, data scientist, analyst.

Want a personalised roadmap or counselling? Contact LessonPlan, Yelahanka, Bangalore.